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11402_1_D1349111.py M11402_2_D1349111.py M X11402_3_D1349111.py M11402_4_D1349111.py M

11402_2_D1349111.py ? ...
Run 24 minutes ago [1 author (You)]
1 import cv2
2 import numpy as np
3
4 Tabbar [Edit | Test | Explain | Document]
5 def onChange(x):
6 pass
7
8 WindowName = 'edge_tool'
9 cv2.namedWindow(WindowName)
10
11 cv2.createTrackbar('Low', WindowName, 50, 255, onChange)
12 cv2.createTrackbar('High', WindowName, 150, 255, onChange)
13 cv2.createTrackbar('Blur_K', WindowName, 1, 15, onChange)
14 cv2.createTrackbar('Method', WindowName, 0, 2, onChange)
15 cv2.createTrackbar('Ksize', WindowName, 0, 5, onChange)
16
17 original_img = cv2.imread('lena.jpg')
18 gray = cv2.cvtColor(original_img, cv2.COLOR_BGR2GRAY)
19
20 while True:
21 low = cv2.getTrackbarPos('Low', WindowName)
22 high = cv2.getTrackbarPos('High', WindowName)
23 blur_val = cv2.getTrackbarPos('Blur_K', WindowName)
24 method = cv2.getTrackbarPos('Method', WindowName)
25
26 ksize_sobel = 2 * cv2.getTrackbarPos('Ksize', WindowName) + 1
27
28 ksize_blur = 2 * blur_val + 1
29 blur_img = cv2.GaussianBlur(gray, (ksize_blur, ksize_blur), 0)
30
31 if method == 0:
32 method_img = cv2.Canny(blur_img, low, high)
33 method_name = "Canny Edge"
34 elif method == 1:
35 sobelx = cv2.Sobel(blur_img, cv2.CV_64F, 1, 0, ksize=ksize_sobel)
36 sobely = cv2.Sobel(blur_img, cv2.CV_64F, 0, 1, ksize=ksize_sobel)
37 abs_sx = cv2.convertScaleAbs(sobelx)
38 abs_sy = cv2.convertScaleAbs(sobely)
39 method_img = cv2.addweighted(abs_sx, 0.5, abs_sy, 0.5, 0)
40 method_name = f"Sobel Edge(k={ksize_sobel})"
41 else:
42 lap = cv2.Laplacian(blur_img, cv2.CV_64F)
43 method_img = cv2.convertScaleAbs(lap)
44 method_name = "Laplacian Edge"
45
46 gray_rainbow = cv2.cvtColor(gray, cv2.COLOR_GRAY2BGR)
47 blur_rainbow = cv2.cvtColor(blur_img, cv2.COLOR_GRAY2BGR)
48 method_rainbow = cv2.cvtColor(method_img, cv2.COLOR_GRAY2BGR)
49
50 cv2.imshow('Original', original_img)
51 cv2.imshow('Gray', gray_rainbow)
52 cv2.imshow('Blur_K (k = %s)' % ksize_blur, blur_rainbow)
53 cv2.imshow('Method: %s' % method_name, method_rainbow)
54 cv2.waitKey(1) <div>edge_tool</div></div><div><div>Low: 50</div><div>High: 150</div><div>Blur_K: 1</div><div>Method: 0</div><div>Ksize: 0</div></div><div><div>Original</div><div>Gray</div><div>Blur_K (k = 1)</div><div>Method: Canny Edge</div></div></div><div><div>PROBLEMS</div><div>OUTPUT</div><div>DEBUG CONSOLE</div><div>TERMINAL</div><div>PORTS</div><div>GIT LENS</div><div>Microsoft Windows [版本 10.0.22000.3246]
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E:\連甲大學\CU 611\大二下\多媒體\期中考程式碼\python -u "E:\連甲大學\CU 611\大二下\多媒體\期中考程式碼\11402_2_D1349111.py"</div><div><div>main</div><div><div>Launchpad</div><div>Run Testcases</div><div><div>0</div><div>0</div></div><div>Live Share</div><div>Git Graph</div><div><div>Disord RPC</div><div><div>Connected to Discord</div></div></div></div>